The Greenhouse Effect

Pupil sheet

Name: _____ **Date:** _____

Safety: No flame in the jar. The box is not waterproof: keep water and vinegar away from it.

1. What I see on the screen

On screen: **CO_2** (ppm), **Temperature** (°C) and **Illuminance** (lx), with charts of the last 30 minutes. I write the temperature down every 10 minutes. **Only the box** goes into the jar; the electronics stay outside.

Be careful what the experiment proves. A closed jar under a lamp warms up *mainly because it has a lid*: the lid stops the warm air leaving. To say anything about CO_2 I need **a control** — a second, identical jar, treated identically, with no extra gas.

Scatter of the rig (empty jars, 15 min) = _____ **°C**
A difference smaller than that *cannot be told apart.*

2. What I do on my own

First I predict — about the *cause*, not the result. The jar will warm up mainly because of:
☐ the lid (it stops convection) ☐ the walls (they absorb light) ☐ the CO_2 (it absorbs and re-emits infrared)

because _____

_____

I check the symmetry before starting. Light: jar A _____ lx, jar B _____ lx. Do they differ by under 5%? ☐ yes ☐ no

Experiment 1 — the lid (lesson 1). Both jars with ordinary air; one with a lid, one without.

Minute	With lid (°C)	No lid (°C)	Difference
0			
10			
20			

Verdict (circle one): detected / borderline / not detected

Experiment 2 — CO_2 (lesson 2). Both with lids. **I make the same gesture in both**: I pour CO_2 from a bottle into one and air from an empty bottle into the other. I wait 10 minutes with the lamp off, then switch it on (minute 0).

CO_2: the gas jar _____ ppm the control _____ ppm

Minute	With CO_2 (°C)	Control (°C)	Difference
0			
10			
20			
25			

Verdict (circle one): detected / borderline / not detected

3. The conclusion I write

I predicted that, and the data show that _____

_____

A garden greenhouse and the atmosphere warm differently, because _____

_____

The very fact that the sensor can measure CO_2 proves that _____

_____

Exit ticket

What warmed the jar, and how do you know?

_____